Total Number of Pages 3

Roll No.....

B.Tech. Electrical Engineering

END SEMESTER EXAMINATION

5th SEMESTER

(Nov. 2024)

EE-301-Power Electronics

Time: 3:00 Hours

Maximum Marks:40

Note: Attempt all parts of the question at one place (Marks may not be awarded otherwise)
Q.1 is COMPULSORY and answer other four from the rest.

1 Draw the waveform/ characteristics depicting the requisite details.

(a) I-V characteristics of Thyristor with depiction of holding current, latching current, forward breakover voltage.

(b) Reverse recovery characteristics of switching power diode and Schottky diode with depiction of I_{RR}, ΔQ_{RR}, and time(s) describing the snappiness factor

(c) Circuit diagram of auxiliary resonant commutation circuit with depiction of polarities on charged capacitor

CO 1,2,5

(d) Current and Voltage waveform both on AC and DC sides of a full thyristorised AC-DC converter feeding a large inductive load with firing angle of near 30°.

(e) Output and current waveform (both on AC and DC sides) of diode rectifier feeding RL load having large inductance substantial source side inductance.

(f) Current and voltage waveform of single-phase AC voltage regulator feeding to purely inductive load (Thyristor Controlled Reactor) for $\alpha > 90°$ (say 120°). Draw the voltage waveform across the switches too.

(g) Waveform of inductor voltage, inductor current and capacitor current of buck convertor with reference to gating pulses.

(h)Waveform for Bipolar pulse width modulation depicting sinusoidal references with triangular carrier and output voltage. **1x8**

2 [a] From the basics, estimate the turn-on losses in the thyristor when connected to a DC source and feeding to a pure resistive load and a R-L (large inductive) load with parallel connected freewheeling diode. **2**

[b] A 400-V, 40-A power MOSFET modulates power from a dc supply of voltage V_s=230V to a load that absorbs an average power of 3 kW. MOSFET data are: C_{iss}=2200pF, C_{oss}=500pF, C_{rss}=180pF, $V_{GS(TH)}$=3V, $R_{DS(ON)}$=0.1Ω, G=9S. The load comprises of a resistance in series with a large inductance. A freewheeling diode is connected across the load to allow the current to be virtually constant. The gate drive consists of a series resistance R_G=22 Ω and a rectangular-unipolar wave voltage supply with V_G =10V operating at a frequency of 25 kHz and a nominal duty cycle m =0.7 (time ratio of positive

CO 1,2

for steady-state switching conditions (both switching & conduction losses). 6

3 [a] Design a turn-off RCD snubber for a power electronic switch feeding an inductive load. If the source voltage V_s=150V DC, load current I_l=10A, and fall time t_f=0.6μs. The switching frequency is f=100kHz, and the duty cycle is D=0.4. Assume the voltage on the switch reaches V_s simultaneous to the current through the switch reaches zero. Further, a time equal to 5-time constants is necessary to discharge the snubber capacitor when the switch is turned ON next. 4 CO 2-3

[b] Ten, 200 V reverse-blocking, ultra fast 35 ns reverse recovery diodes are to be employed in series in a 1500 V peak, string voltage application. If the maximum device reverse leakage current is 10 mA calculate for worst case conditions, the maximum value of sharing resistance(*commercially available resistance*) and the total power dissipation during blocking for static balancing. Further, for dynamic balancing of the string, estimate the minimum possible sharing capacitance for a worst case reverse recovery conditions occurring at maximum junction temperature, di/dt, and a value of Q=6μC. Estimate the energy loss due to capacitor discharge. 4

4 [a] A purely resistive load is supplied from an ideal single-phase supply of fixed voltage and frequency AC source, through an ideal diode bridge rectifier with an ideal SCR switch on the DC side connecting it in series with the load. The switch is gated synchronously during every half cycle at an angle α. Sketch the supply current waveform and the load current waveform for a switching angle α=60. Derive an expression for the rms value of the load current. 3 CO 2,3,4

[b] Determine the average value of the load current drawn by a *1Ph* thyristorised rectifier fed from 230V,50Hz AC source having source inductance (L_s) of 11mH, when thyristors are fired at $\alpha = \pi/3$. Assume $\omega L >> R$, to affirm constant current loading, while supplying to connected load(R=10 Ω, L large value). 2

[c] A full controlled thyristorised AC-DC converter fed from 230V, 50Hz supply feeds to RL load (R=12Ω, L=100mH). For the firing angle α=60°, determine the power absorbed by the load and peak-peak current ripple. Verify whether the current is continuous or discontinuous. 3

5 [a] A step-down dc converter is used to convert a 100-V dc supply to 75-V output. The inductor is 200 μH with a resistive load 2.2 Ω. If the MOSFET power switch has an on time of 50 μs and conduction is continuous, calculate the rms values of the diode, switch, and load currents. 4 CO 2,3,4

[b] Design a boost converter having following specifications for CCM mode: V_{in}=48V[40V-75V]; V_{out}=110V; P_{out}=900W[150W-1200W]; f_s=100kHz, η=85%. 4

CO 2-

determine whether the current is continuous or discontinuous, the output power, rms current, and input power factor for $\alpha = \pi/6$.

[b] Discuss in brief details with circuit diagrams, waveforms the single phase to single phase, three phase to single phase cycloconverter. Compare the operation of single phase to single phase cycloconverter with dual converter.

OR

Discuss in brief details with waveforms the Bipolar and Unipolar Sinusoidal Pulse Width Modulation (SPWM) for a single phase bridge inverter. Compute the output of the inverter for Bipolar and Unipolar SPWM at fundamental frequency. Comment on the harmonic spectrum of the two schemes. 4

7 [a] Derive the expression for the load voltages (v_{an}, v_{bn}, v_{cn}) of the three phase thyristorised inverter for 180° conduction. Draw a neat diagram to support the analysis. 4

[b] A single phase full bridge Inverter feeding a RL load(L=30mH, R=10Ω) with square and quasi-square voltage. If V_{DC}=150V and f=50 Hz, compare the total harmonic distortions of the voltage for square and quasi-square voltage (*where pulse is chopped off symmetrically in each half & on each side by 30°*). 4 CO 2-5

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FIFTH SEMESTER B.TECH. ELECTRICAL ENGINEERING EXAMINATION, NOV, 2024

END SEMESTER

EE-303, POWER TRANSMISSION AND DISTRIBUTION

Time: 03:00Hours

Max.Marks:40

Note :Attempt any five questions.
All questions carry equal marks. Assume suitable missing data, if any.

1. Explain briefly: CO1 L4 2x4
   (a) Compare underground and overhead transmission systems
   (b) Discuss why series compensation increases the power transmission capability.
   (c) Explain reason for Ferranti effect to be evident on long lines
   (d) Skin effect and proximity effect
2. (a) Compute the flux linkages and the inductance for three phase line with equilateral spacing. CO2 L2 4
   (b) A 220kV, 50 Hz, 200km three phase line has conductors at the corners of a triangle with asymmetrical spacing of 6m, 6m, 12m. The conductor radius is 1.81cm. Compute the capacitance/ phase/km, capacitive reactance and charging current. CO2 L5 4
3. (a) Represent the nominal pi circuit of a medium length transmission line and draw itsphasor diagram. Compute its ABCD parameters. CO4 L2 4
   (b) A three phase 200km long high voltage line has Z=14.1+51.48 ohms. $y=j1.194 \times 10^{-3}$ siemens. Find the following (i) characteristic impedance (ii) propagation constant (iii) Parameter A (iv)Parameter C. CO4 L3 4
4. (a) Discuss factors and conditions influencing corona loss.How can this loss be reduced practically? CO3 L4 4
   (b) Compute (i) critical disruptive voltage (ii) visual critical voltage for a three phase line which is 160km long, diameter 1.036cm, 2.44m delta spacing, air temperature is 26.6°C, barometric pressure of 73.15cm of mercury, operating at voltage of 110kV at 50 Hz. Assume surface irregularity factor of 0.85 and m_v=0.72. CO3 L5 4
5. (a) Explain the need of grading of cables. Analyze the capacitive grading method considering three layers of dielectrics having different permittivities, CO3 L3 4
   (b) An insulator string designed for 66kV, has four discs. Assume the shunt capacitance between each joint and metal work is 10% of the capacitance of each disc. Find how the voltage is distributed across different discs and the string efficiency. CO1 L3 4
6. (a) An overhead line is supported between two supports A and B at the same level. Discuss the computation of sag and develop a suitable expression.[CO#2] [BTL#L3]
   (b) 230kV transmission line has parameters: weight of conductor=680kg/km. length of span=260m, tension=3000kg (maximum), factor of safety is 5. Find height above the ground at which the conductor needs to be supported if the required clearance from ground is 12m. CO2 L5 4
7. Write short notes on any two of the following: CO5 L4 4x2
   (i) Design of Lightning arrestors
   (ii) Importance of High Voltage DC Transmission systemsand its main configurations
   (iii) Layout of power system network

distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the centre of each cluster, respectively. Use the K-Means algorithm to show the three cluster centres after the first round of execution. Also find out the final three clusters. [5] [CO5] [L3]

Total no. of Pages: 04 Roll no.....

V & VII SEMESTER

B.Tech

END TERM EXAMINATION *Dec-2024*

EE323 FUNDAMENTALS OF MACHINE LEARNING

Time: 03:00 Hours *Max. Marks: 40*

Note : All questions carry equal marks (10 marks).
Attempt "ANY 4" questions
Assume suitable missing data, if any.

Q.1

A. What is the output of the following code. Also write the complete steps involved. [4][CO2][L2]

```
def iChooseYou(pc):
    pc.append("Kamal")
    pc.sort()

faculty = ["Pankaj", "Neeraj"]
iChooseYou(faculty)
print(faculty)
```

B. Consider the following optimization problem:

minimize $f(x) = x^3 - 3x^2 + 2x + 1$, subject to $-1 \le x \le 2$

(i) Test whether f(x) is a convex function. [2] [CO1] [L4]
(ii) Find the optimal solution x* that minimizes f(x) over the given interval. [2] [CO1] [L3]
(iii) Find the minimum value of f(x) over the given interval. [2] [CO1] [L3]

Q.2 [Linear Regression + GD Optimization]

A. Suppose we have a data set of 100 points whose first few rows are shown below, and that we'd like to predict $\vec{y}$ from $\vec{v}$ and $\vec{w}$. Suppose we create a design matrix **X** whose first column is $\vec{v}$, second column is $\vec{w}$, and third column $\vec{u}$ is a new feature $u_i = |v_i|$. The resulting model is $\hat{y}_i = \beta_1 v_i + \beta_2 w_i + \beta_3 u_i$. **The top row is row 1**, e.g. $y_1 = 4$.

y	v	w
4	-30	1
6	-40	2
5	20	3

i. For the data above, suppose we arbitrarily pick $\vec{\beta} = [0.1, 12, 0.2]^T$. What is $\hat{y}_1$? **[1] [CO4] [L3]**

ii. For the data above, let $\vec{e}$ be the residual vector if $\vec{\beta} = [0.1, 12, 0.2]^T$. What is $|e_1|$? **[1] [CO4] [L3]**

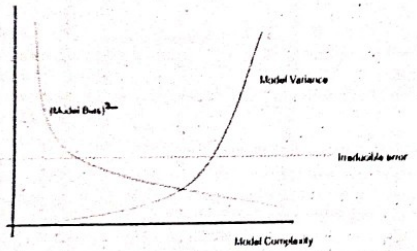
iii. Suppose that you have m = 14 training examples with n = 3 features. The normal equation is $\theta = (X^TX)^{-1}X^Ty$. For the given values of m and n, what are the dimensions of θ, X, and y? Show the calculations. **[3] [CO4] [L2]**

B. Write short notes on Gradient Descent Algorithm? Explain with a pseudo code? Now, consider the function $f(x, y) = x^2 + 2y^2$. Find the minimum value of $f(x, y)$ using gradient descent algorithm with a learning rate of 0.1 and starting point $(x, y) = (1, 1)$. Show two iterations. **[5] [CO4] [L2]**

Q.3 [Bias Variance trade off + Confusion Matrix]

A. What is meant by overfitting and underfitting of data with examples? **[4] [CO3] [L2]**

B. Sketch the MSE (Mean Squared Error) on the graph below. Where does its minimum occur? Draw a star on your MSE plot where the minimum occurs. Elaborate your answer. **[2] [CO3] [L3]**



C. For a C-class problem, the confusion matrix is a C×C matrix that tells us, f any two classes c, c' ∈ [C], how many instances of class c were classified a c' by the model. In the example below, C=2, there were P+Q+R+S points i the test set where P, Q, R, S are strictly positive integers. Give expressions for the specified quantities in terms of P, Q, R, S. No derivations needed. Note that y denotes the true class of a test point and $\hat{y}$ is the predicted class for that point. **[4] [CO3] [L4]**

		Predicted class $\hat{y}$	
		+1	−1
True class y	+1	P	Q
	−1	R	S

Confusion Matrix

i. $P[\hat{y} = y \mid \hat{y} = 1]$
ii. $P[\hat{y} = 1 \mid y = 1]$
iii. $P[\hat{y} \neq y \mid \hat{y} = 1]$
iv. $P[y = \hat{y}]$

Q.4 [Dimensionality Reduction]

A. Explain why dimensionality reduction is useful in machine learning? What are the commonly used dimensionality reduction techniques in machine learning? Explain any one in brief. **[5] [CO4] [L2]**

B. Determine the Singular value decomposition (SVD) of the following 3×3 matrix "A"

$$A = \sqrt{2}\begin{bmatrix} 0 & 1 & 1 \\ 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Also, comment on how the eigenvalues and singular values are related? **[5] [CO5] [L5]**

Q.5 [Clustering]

A. Explain k-means algorithm with a pseudo code. Also, discuss the effects of outliers and initial seed points on the convergence of k-means clustering algorithm with suitable examples? **[5] [CO5] [L2]**

B. Suppose that the data mining task is to cluster the following eight points (with x, y representing location) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4,9). The

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V / VII SEMESTER B.Tech. [EE]

END-SEM EXAMINATION *November-2024*

EE-325 INTERNET OF THINGS

Time: 3 Hour *Max. Marks: 50*

Note: Attempt any five questions. All questions carry equal marks

Q.1[a] Compare wireless sensor network versus IoT. [3] **CO2 L2**

[b] Discuss the IoT stack in detail with examples for each layer. [7] **CO1, CO4 L2**

Q.2[a] Explain the working of MQ-02 sensor and its interfacing with the Arduino microcontroller. How is MQ-05 different from MQ-02. [5] **CO4 L2**

[b] Explain the MQTT scheme: publish – subscribe pattern. Also explain the three levels QoS supported by MQTT. [5] **CO2 L2**

Q.3[a] Discuss the various types of CoAP – request – response layer messages. [4] **CO2 CO5 L2**

[b] Discuss the various types of transport protocols. [6] **CO2 CO5 L3**

Q.4[a] One of the PC on DTU network shows the IP address as 10.13.10.94. Which version of IP address is this? What should be its subnet mask? What is the class of this IP address and justify the use of this class of IP address for DTU network. [6] **CO3 L2 L3**

[b] Discuss IPV6 packet format. [4] **CO3 L3**

Q.5[a] Discuss and compare the three deployment models related to cloud computing. [5] **CO2 L2 L3**

[b] Enlist and discuss in brief the challenges related to IoT with cloud [5] **CO5 L3**

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Q.6 [a] Discuss the various criteria one should consider while selecting a cloud service provider. [5] **CO5 L2 L3**

[b] Discuss the various types of machine learning models. [5] **CO5 L2 L3**

Q.7[a] Discuss how IoT can be used for smart warehouse monitoring. [5] **CO6 L2 L3**

[b] How IoT can be used for water quality monitoring? Discuss the end-to-end solution. [5] **CO6 L2 L3**

Q.8 Write short notes on any two. **[2x 5] All CO's**

i. Machine learning.
ii. Components of BLE
iii. Uniform resource identifier
iv. Difference between cloud and fog computing.
v. Service field in IPV4

*****End****

7 b A contract has been awarded the contract to construct a 6-mile long tunnel in the mountain. During the five years construction period, the contractor will need water from a nearby stream. He will construct a pipeline to carry the water to the main construction yard. An analysis of costs for various pipe sizes is as follows: [5][CO# 4]

	Pipe size (cm)			
	6	8	10	15
Installed cost of pipeline and pump(in ₹)	22,000	23,000	25,000	30,000
Cost/hour for pumping (in ₹)	1.20	0.65	0.50	0.40

At the end of 5 years, the pipe and pump will have a salvage value equal to the cost of removing them. The pump will operate 2000 hours per year. The lowest interest rate at which the contractor is willing to invest money is 7%. (The minimum required interest rate for invested money is called the minimum attractive rate of return(MARR)). Select the alternative with the least present worth of cost

Total no. of Pages : 04 *Roll no.....*

SEMESTER-V

B.Tech

END TERM EXAMINATION *Nov/Dec-2024*

HU-301 Engineering Economics

Time: 03:00 Hours *Max. Marks: 50*

Note : Attempt any five questions.
All questions carry equal marks.
Assume suitable missing data, if any.

1 a Define Sustainable Development. What is the role of an Engineer in promoting Sustainable Development? Discuss. [2+3=5] [CO# 1+3]

1 b Mr. A furnishes the following data relating to the manufacturing of a Standard product during the month of January 2024. [2+3=5] [CO# 4]

You are required to prepare

1) Cost Sheet

2) A Statement showing Profit for the Period.

Particular	Price/ Cost /Time
Raw material purchased	₹ 15000
Opening Stock of Raw Materials	₹ 4000
Closing Stock of Raw Materials	₹ 5000
Direct Labour Cost	₹ 9000
Machine Hours Worked	900 Hrs
Machine Hours Rate	₹ 5/ Hour
Carriage Inwards	₹ 1000
Administrative Overheads	20% Of Work Cost
Selling Overheads	₹0.50 Per Unit Sold
Units produced	17100 Units
Opening stock of finished goods	2000 Units @ ₹ 4.50 Per Unit
Units Sold	16000 Units
Selling Price per Unit	₹ 14

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2 a Differentiate between Microeconomics and Macroeconomics with suitable examples [5][CO# 2]

2 b Summit has ₹90 with him. He intends to purchase goods X and Y with his money. The market price of X and Y per unit is ₹10. The marginal utility schedule of goods X and Y is given below. Find out how many units of X and Y should summit purchase so that he gets maximum satisfaction. [5] [CO# 2]

Units of Commodity	1	2	3	4	5	6	7	8	9	10
MUx	80	72	64	56	48	40	32	24	16	8
MUy	40	32	24	20	16	12	8	4	0	0

3 a What do you mean by Monopolistic Competition? How it is different from Perfect Competition? Discuss with suitable examples. [2+3=5] [CO# 1&3]

b A firm is borrowing 100 lakhs/ year for three years for technological upgradation. The loan will be repaid two years later@ 15%/ year. Calculate how much will be repaid. [5][CO# 2]

4 a What do you mean by an Opportunity Cost? How is it different from the Actual cost? Discuss with suitable examples . [2+3=5] [CO# 1&3]

b Calculate Total Fixed Cost (TFC), Total Variable Cost (TVC), Average Total Cost (ATC), Average Fixed Cost (AFC) and Marginal Cost (MC) from the following: 5 [CO# 2]

Output (in units)	0	1	2	3	4	5	6
TC (₹)	40	100	120	130	150	190	240

5 a What do you mean by Monetary Policy? How monetary policy may be used for controlling Inflation? Discuss [2+3=5] [CO# 1&3]

b A man owns a corner plot. He must decide which of the two alternatives to select in trying to obtain a desirable return on his investment. After much study and calculation, he decides that there are two best alternatives. Bank interest rate is 12% compounded annually. Details are given below: [5][CO# 2]

	Build Gas station	Build Ice-cream stand
Initial Investment (in ₹)	20,00,000	36,00,000
Annual Property Tax (in ₹)	80,000	1,50,000
Annual Income (in ₹)	8,00,000	9,80,000
Life of Building (in Years)	20	20
Salvage Value (value which owner may receive after completing life) (in ₹)	0	0

Evaluate the alternative based on the future worth method.

6 a What do you mean by Fiscal Policy? Suggest a fiscal policy to control air pollution in Delhi. [2+3=5] [CO# 1&3]

b A company invests in one of the two mutually exclusive alternatives. The life of both alternatives is estimated to be 5years with the following initial investments, annual return and salvage value (money received after selling the machine at the end of its operational life) [5][CO# 2]

	Alternative A	Alternative B
Initial Investment (in ₹)	1,50,000	1,75,000
Annual Equal Return (in ₹)	60,000	70,000
Salvage value (in ₹)	15000	35000

Determine the best alternative based on the annual equivalent method when rate of interest is 25%.

7 a You are CEO of an electronic company. Now, you want to expand your business in a foreign country. What are the factors you will consider while deciding foreign location for your company? [5][CO# 4]

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